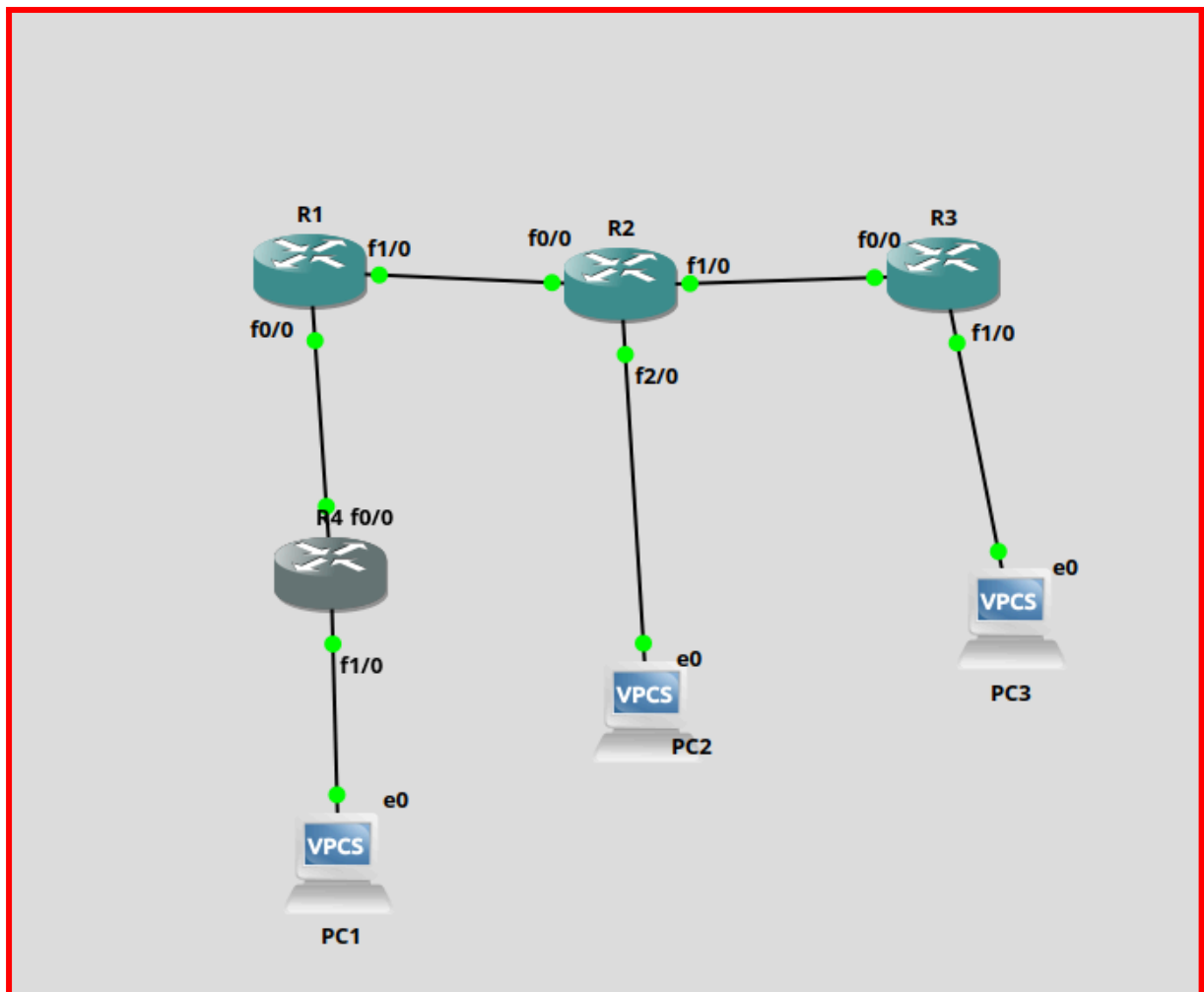




Configuration Ip des 3 pc

PC1

nginx

[Copier le code](#)

```
ip 10.10.10.10 255.255.255.0 10.10.10.1
```

PC2

nginx

[Copier le code](#)

```
ip 10.20.20.10 255.255.255.0 10.20.20.1
```

PC3

nginx

[Copier le code](#)

```
ip 172.16.1.10 255.255.255.0 172.16.1.1
```

Configuration IP Routers

R1

```
hostname R1
interface f0/0
ip address 172.16.2.1 255.255.255.0
no shut
interface f1/0
ip address 192.168.1.1 255.255.255.0
no shut
```

R2

```
hostname R2
interface f0/0
ip address 192.168.1.2 255.255.255.0
no shut
interface f1/0
ip address 192.168.2.1 255.255.255.0
no shut
interface f2/0
ip address 10.20.20.1 255.255.255.0
no shut
```

R3

```
hostname R3
interface f0/0
ip address 192.168.2.2 255.255.255.0
no shut
interface f1/0
ip address 172.16.1.1 255.255.255.0
no shut
```

R4

```
hostname R4
interface f0/0
ip address 172.16.2.2 255.255.255.0
no shut
interface f1/0
ip address 10.10.10.1 255.255.255.0
no shut
```

ROUTAGE STATIQUE – CONFIGURATION COMPLÈTE

R1

```
R1(config)#ip route 10.10.10.0 255.255.255.0 172.16.2.2
R1(config)#ip route 10.20.20.0 255.255.255.0 192.168.1.2
R1(config)#ip route 172.16.1.0 255.255.255.0 192.168.1.2
R1(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
R1(config)#
```

```
ip route 10.10.10.0 255.255.255.0 172.16.2.2      ! Vers R4 → PC1
ip route 10.20.20.0 255.255.255.0 192.168.1.2    ! Vers R2 → PC2
ip route 172.16.1.0 255.255.255.0 192.168.1.2    ! Vers R2 → PC3
ip route 192.168.2.0 255.255.255.0 192.168.1.2    ! Réseau R2-R3 via R2
```

R2

```
R2#CONF T
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.10.10.0 255.255.255.0 192.168.1.1
R2(config)#ip route 172.16.2.0 255.255.255.0 192.168.1.1
R2(config)#ip route 172.16.1.0 255.255.255.0 192.168.2.2
R2(config)#
```

```
ip route 10.10.10.0 255.255.255.0 192.168.1.1    ! R1 → R4 → PC1
ip route 172.16.2.0 255.255.255.0 192.168.1.1    ! Réseau R1-R4
ip route 172.16.1.0 255.255.255.0 192.168.2.2    ! Vers R3 → PC3
```

R3

```
R3(config)#ip route 10.10.10.0 255.255.255.0 192.168.2.1
R3(config)#ip route 10.20.20.0 255.255.255.0 192.168.2.1
R3(config)#ip route 172.16.2.0 255.255.255.0 192.168.2.1
R3(config)#
```

```
ip route 10.10.10.0 255.255.255.0 192.168.2.1    ! Vers R2 → R1 → R4 → PC1
ip route 10.20.20.0 255.255.255.0 192.168.2.1    ! Vers PC2
ip route 172.16.2.0 255.255.255.0 192.168.2.1    ! Réseau R1-R4
```

R4

```
R4(config)#ip route 192.168.1.0 255.255.255.0 172.16.2.1
R4(config)#ip route 192.168.2.0 255.255.255.0 172.16.2.1
R4(config)#ip route 10.20.20.0 255.255.255.0 172.16.2.1
R4(config)#ip route 172.16.1.0 255.255.255.0 172.16.2.1
R4(config)#
```

```
ip route 192.168.1.0 255.255.255.0 172.16.2.1      ! Vers R1
ip route 192.168.2.0 255.255.255.0 172.16.2.1      ! R3 via R1 → R2
ip route 10.20.20.0 255.255.255.0 172.16.2.1       ! PC2 via R1 → R2
ip route 172.16.1.0 255.255.255.0 172.16.2.1       ! PC3 via R1 → R2 → R3
```

TEST PING pC1

```
PC1>
PC1> ping 10.20.20.10

10.20.20.10 icmp_seq=1 timeout
10.20.20.10 icmp_seq=2 timeout
10.20.20.10 icmp_seq=3 timeout
84 bytes from 10.20.20.10 icmp_seq=4 ttl=61 time=58.763 ms
84 bytes from 10.20.20.10 icmp_seq=5 ttl=61 time=56.132 ms
:

PC1> ping 172.16.1.10

172.16.1.10 icmp_seq=1 timeout
172.16.1.10 icmp_seq=2 timeout
84 bytes from 172.16.1.10 icmp_seq=3 ttl=60 time=63.274 ms

```

TEST PING pC2

```
PC2>  
PC2>  
PC2> ping 10.10.10.10  
  
84 bytes from 10.10.10.10 icmp_seq=1 ttl=61 time=32.840 ms  
84 bytes from 10.10.10.10 icmp_seq=2 ttl=61 time=37.209 ms  
84 bytes from 10.10.10.10 icmp_seq=3 ttl=61 time=36.915 ms  
84 bytes from 10.10.10.10 icmp_seq=4 ttl=61 time=36.816 ms  
^C  
PC2> ping 172.16.1.10  
  
84 bytes from 172.16.1.10 icmp_seq=1 ttl=62 time=23.651 ms  
84 bytes from 172.16.1.10 icmp_seq=2 ttl=62 time=26.171 ms  
84 bytes from 172.16.1.10 icmp_seq=3 ttl=62 time=26.764 ms  
^C
```

TEST PING pc3

```
PC3> ping 10.10.10.10

84 bytes from 10.10.10.10 icmp_seq=1 ttl=60 time=48.798 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=60 time=46.651 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=60 time=47.594 ms
84 bytes from 10.10.10.10 icmp_seq=4 ttl=60 time=46.794 ms
84 bytes from 10.10.10.10 icmp_seq=5 ttl=60 time=46.677 ms
^C
PC3> 10.20.20.10
Bad command: "10.20.20.10". Use ? for help.

PC3> ping 10.20.20.10

84 bytes from 10.20.20.10 icmp_seq=1 ttl=62 time=30.829 ms
84 bytes from 10.20.20.10 icmp_seq=2 ttl=62 time=26.924 ms
84 bytes from 10.20.20.10 icmp_seq=3 ttl=62 time=26.626 ms
^C
PC3> █
```

Reconfigurer le réseau en utilisant le protocole de routage

Il faut d'abord supprimer

R1

```
R1(config)#no ip route 10.10.10.0 255.255.255.0 172.16.2.2
R1(config)#no ip route 10.20.20.0 255.255.255.0 192.168.1.2
R1(config)#no ip route 172.16.1.0 255.255.255.0 192.168.1.2
R1(config)#no ip route 192.168.2.0 255.255.255.0 192.168.1.2
R1(config)#█
```

R2

```
R2(config)#no ip route 10.10.10.0 255.255.255.0 192.168.1.1
R2(config)#no ip route 172.16.2.0 255.255.255.0 192.168.1.1
R2(config)#no ip route 172.16.1.0 255.255.255.0 192.168.2.2
R2(config)#█
```

Même pour les deux autre

CONFIGURATION RIP v2

R1

```
R1(config)#router rip
R1(config-router)# version 2
R1(config-router)# no auto-summary
R1(config-router)# network 172.16.2.0
R1(config-router)# network 192.168.1.0
R1(config-router)#
R1(config-router)#end
R1#
```

```
router rip
version 2
no auto-summary
network 172.16.2.0
network 192.168.1.0
```

R2

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 192.168.1.0
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.20.20.0
R2(config-router)#
```

```
router rip
version 2
no auto-summary
network 192.168.1.0
network 192.168.2.0
network 10.20.20.0
```

R3

```
R3(config)#router rip
R3(config-router)# version 2
R3(config-router)# no auto-summary
R3(config-router)# network 192.168.2.0
R3(config-router)# network 172.16.1.0
R3(config-router)#
R3(config-router)#end
```

R4

```
R4(config)#router rip
R4(config-router)# version 2
R4(config-router)# no auto-summary
R4(config-router)# network 172.16.2.0
R4(config-router)# network 10.10.10.0
R4(config-router)#
```

```
router rip
  version 2
  no auto-summary
  network 172.16.2.0
  network 10.10.10.0
```

Teste ping

PC1

```
PC1>
PC1> ping 10.20.20.10

84 bytes from 10.20.20.10 icmp_seq=1 ttl=61 time=50.040 ms
84 bytes from 10.20.20.10 icmp_seq=2 ttl=61 time=37.061 ms
^C
PC1> ping 172.16.1.10

84 bytes from 172.16.1.10 icmp_seq=1 ttl=60 time=56.370 ms
84 bytes from 172.16.1.10 icmp_seq=2 ttl=60 time=46.559 ms
^C
PC1> 
```

PC2

```
PC2>
PC2> ping 10.10.10.10

84 bytes from 10.10.10.10 icmp_seq=1 ttl=61 time=35.545 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=61 time=36.434 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=61 time=36.249 ms
^C
PC2> ping 172.16.1.10

84 bytes from 172.16.1.10 icmp_seq=1 ttl=62 time=28.723 ms
84 bytes from 172.16.1.10 icmp_seq=2 ttl=62 time=26.413 ms
^C
PC2> 
```

PC3

```
PC3>
PC3> ping 10.10.10.10
84 bytes from 10.10.10.10 icmp_seq=1 ttl=60 time=50.531 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=60 time=46.477 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=60 time=46.521 ms
^C
PC3> ping 10.20.20.10
84 bytes from 10.20.20.10 icmp_seq=1 ttl=62 time=24.120 ms
84 bytes from 10.20.20.10 icmp_seq=2 ttl=62 time=26.417 ms
84 bytes from 10.20.20.10 icmp_seq=3 ttl=62 time=26.596 ms
^C
PC3> 
```

Remplacer RIP par OSPF dans une seule aire (Area 0)

SUPPRIMER RIP SUR TOUS LES ROUTEURS

no router rip ==> pour les 4 routeurs

CONFIGURER OSPF (AREA 0)

R1

```
R1(config)#no router rip
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 172.16.2.0 0.0.0.255 area 0
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#
```

```
router ospf 1
  router-id 1.1.1.1
  network 172.16.2.0 0.0.0.255 area 0
  network 192.168.1.0 0.0.0.255 area 0
```

R2

```
R2(config)#router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 192.168.1.0 0.0.0.255 area 0
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.20.20.0 0.0.0.255 area 0
R2(config-router)#
*Mar  1 00:36:32.443: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1.
from LOADING to FULL, Loading Done
R2(config-router)#
```

```
router ospf 1
  router-id 2.2.2.2
  network 192.168.1.0 0.0.0.255 area 0
  network 192.168.2.0 0.0.0.255 area 0
  network 10.20.20.0 0.0.0.255 area 0
```

R3

```
router ospf 1
  router-id 3.3.3.3
  network 192.168.2.0 0.0.0.255 area 0
  network 172.16.1.0 0.0.0.255 area 0
```

```
R3(config)#router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 192.168.2.0 0.0.0.255 area 0
R3(config-router)# network 172.16.1.0 0.0.0.255 area 0
R3(config-router)#
*Mar  1 00:37:00.707: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0
from LOADING to FULL, Loading Done
R3(config-router)#
```

R4

```
% Ambiguous command: no router rip
R4(config)#no router rip
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 172.16.2.0 0.0.0.255 area 0
R4(config-router)# network 10.10.10.0 0.0.0.255 area 0
R4(config-router)#
*Mar  1 00:37:09.299: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 o
```

```

router ospf 1
  router-id 4.4.4.4
  network 172.16.2.0 0.0.0.255 area 0
  network 10.10.10.0 0.0.0.255 area 0

```

Commande pour verification les route ospf

- show ip route ospf
- show ip ospf neighbor

```

R1#sh ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:34    192.168.1.2  FastEthernet1/0
4.4.4.4          1     FULL/BDR        00:00:31    172.16.2.2   FastEthernet0/0
R1#show ip route ospf
  172.16.0.0/24 is subnetted, 2 subnets
O       172.16.1.0 [110/3] via 192.168.1.2, 00:02:37, FastEthernet1/0
  10.0.0.0/24 is subnetted, 2 subnets
O       10.20.20.0 [110/2] via 192.168.1.2, 00:02:37, FastEthernet1/0
O       10.10.10.0 [110/2] via 172.16.2.2, 00:02:37, FastEthernet0/0
O      192.168.2.0/24 [110/2] via 192.168.1.2, 00:02:37, FastEthernet1/0
R1#show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:39    192.168.1.2  FastEthernet1/0
4.4.4.4          1     FULL/BDR        00:00:37    172.16.2.2   FastEthernet0/0
R1#

```

==> ca ping aussi

CONFIGURATIONS eBGP

R1 (R1 — AS 65101)

```

router bgp 65101
  neighbor 172.16.2.2 remote-as 65100
  neighbor 192.168.1.2 remote-as 65102

  network 172.16.2.0 mask 255.255.255.0
  ʹʹ Demander à ChatGPT mask 255.255.255.0

```

```

R1(config)#router bgp 65101
R1(config-router)# neighbor 172.16.2.2 remote-as 65100
R1(config-router)# neighbor 192.168.1.2 remote-as 65102
R1(config-router)#
R1(config-router)# network 172.16.2.0 mask 255.255.255.0
R1(config-router)# network 192.168.1.0 mask 255.255.255.0
R1(config-router)#

```

R2(AS 65102)

```

router bgp 65102
 neighbor 192.168.1.1 remote-as 65101
 neighbor 192.168.2.2 remote-as 65103

 network 10.20.20.0 mask 255.255.255.0
 network 192.168.1.0 mask 255.255.255.0
 network 192.168.2.0 mask 255.255.255.0

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router bgp 65102
R2(config-router)# neighbor 192.168.1.1 remote-as 65101
R2(config-router)# neighbor 192.168.2.2 remote-as 65103
R2(config-router)#
R2(config-router)# network 10.20.20.0 mask 255.255.255.0
R2(config-router)# network 192.168.1.0 mask 255.255.255.0
R2(config-router)# network 192.168.2.0 mask 255.255.255.0
R2(config-router)#

```

R3 (— AS 65103)

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 65103
R3(config-router)# neighbor 192.168.2.1 remote-as 65102
R3(config-router)#
R3(config-router)# network 172.16.1.0 mask 255.255.255.0
R3(config-router)# network 192.168.2.0 mask 255.255.255.0
R3(config-router)#

```

```
router bgp 65102
  neighbor 192.168.1.1 remote-as 65101
  neighbor 192.168.2.2 remote-as 65103

  network 10.20.20.0 mask 255.255.255.0
  network 192.168.1.0 mask 255.255.255.0
  network 192.168.2.0 mask 255.255.255.0
```

R4(AS 65100)

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router bgp 65100
R4(config-router)# neighbor 172.16.2.1 remote-as 65101
R4(config-router)#
R4(config-router)# network 10.10.10.0 mask 255.255.255.0
R4(config-router)# network 172.16.2.0 mask 255.255.255.0
R4(config-router)#
```

```
router bgp 65100
  neighbor 172.16.2.1 remote-as 65101

  network 10.10.10.0 mask 255.255.255.0
  network 172.16.2.0 mask 255.255.255.0
```

Teste ping

```
PC1>
PC1> ping 10.20.20.10

84 bytes from 10.20.20.10 icmp_seq=1 ttl=61 time=49.606 ms
84 bytes from 10.20.20.10 icmp_seq=2 ttl=61 time=36.652 ms
84 bytes from 10.20.20.10 icmp_seq=3 ttl=61 time=36.273 ms
^C
PC1> ping 172.16.1.10

84 bytes from 172.16.1.10 icmp_seq=1 ttl=60 time=60.466 ms
84 bytes from 172.16.1.10 icmp_seq=2 ttl=60 time=46.442 ms
84 bytes from 172.16.1.10 icmp_seq=3 ttl=60 time=47.065 ms
^C
PC1>
```

eBGP entre R3 ET R2 + iBGP entre R1–R2–R4

Faut d'abord supprimer l'autre route

no router bgp 65101

R1

```
R1(config)#no router bgp 65101
R1(config)#router bgp 65101
R1(config-router)# neighbor 192.168.1.2 remote-as 65101 ! iBGP vers R2
R1(config-router)# neighbor 172.16.2.2 remote-as 65101 ! iBGP vers R4
R1(config-router)#
R1(config-router)# network 172.16.2.0 mask 255.255.255.0
R1(config-router)# network 192.168.1.0 mask 255.255.255.0
R1(config-router)#
*Mar 1 00:53:01.887: %BGP-5-ADJCHANGE: neighbor 172.16.2.2 Down BGP protocol in
ititialization
*Mar 1 00:53:01.887: %BGP-5-ADJCHANGE: neighbor 192.168.1.2 Down BGP protocol i
nitialization
R1(config-router)#
```

```
router bgp 65101
 neighbor 192.168.1.2 remote-as 65101 ! iBGP vers R2
 neighbor 172.16.2.2 remote-as 65101 ! iBGP vers R4

 network 172.16.2.0 mask 255.255.255.0
 network 192.168.1.0 mask 255.255.255.0
```

R2

```
R2(config)#no router bgp 65102
R2(config)#
*Mar 1 00:53:24.411: %BGP-5-ADJCHANGE: neighbor 192.168.2.2 Down BGP protocol i
nitialization
R2(config)#router bgp 65101
R2(config-router)# neighbor 192.168.1.1 remote-as 65101 ! iBGP R1
R2(config-router)# 172.16.2.2 remote-as 65101 ! iBGP R4 (via R1 path)
R2(config-router)# neighbor 192.168.2.2 remote-as 65103 ! eBGP R3
R2(config-router)#
R2(config-router)# network 192.168.1.0 mask 255.255.255.0
R2(config-router)# network 192.168.2.0 mask 255.255.255.0
R2(config-router)# network 10.20.20.0 mask 255.255.255.0
R2(config-router)#
```

```

router bgp 65101
  neighbor 192.168.1.1 remote-as 65101    ! iBGP R1
  neighbor 172.16.2.2 remote-as 65101    ! iBGP R4 (via R1 path)
  neighbor 192.168.2.2 remote-as 65103    ! eBGP R3

  network 192.168.1.0 mask 255.255.255.0
  network 192.168.2.0 mask 255.255.255.0
  network 10.20.20.0 mask 255.255.255.0

```

R3

```

Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
*Mar  1 00:53:38.283: %BGP-3-NOTIFICATION: sent to neighbor 192.168.2.1 2/2 (peer in wrong AS) 2 bytes FE4D
R3(config)# FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF 002D 0104 FE4D 00B4 C0A8 0201 1002 0601 0400 0100 0102 0280 0002 0202 00
R3(config)#no router bgp 65103
R3(config)#router bgp 65101
R3(config-router)# neighbor 192.168.2.1 remote-as 65101
R3(config-router)#
R3(config-router)# network 172.16.1.0 mask 255.255.255.0
R3(config-router)# network 192.168.2.0 mask 255.255.255.0
R3(config-router)#

```

```

router bgp 65103
  neighbor 192.168.2.1 remote-as 65101

  network 172.16.1.0 mask 255.255.255.0
  network 192.168.2.0 mask 255.255.255.0

```

R4

```

R4(config-router)#
*Mar  1 00:53:32.151: %BGP-3-NOTIFICATION: received from neighbor 172.16.2.1 2/2 (peer in wrong AS) 2 bytes FE4C
R4(config-router)#exit
R4(config)#no router bgp 65100
R4(config)#
R4(config)#router bgp 65101
R4(config-router)# neighbor 172.16.2.1 remote-as 65101    ! iBGP vers R1
R4(config-router)# 192.168.1.2 remote-as 65101    ! iBGP vers R2 (via R1 path)
R4(config-router)#
R4(config-router)# network 172.16.2.0 mask 255.255.255.0
R4(config-router)# network 10.10.10.0 mask 255.255.255.0
R4(config-router)#

```

```
router bgp 65101
  neighbor 172.16.2.1 remote-as 65101    ! iBGP vers R1
  neighbor 192.168.1.2 remote-as 65101  ! iBGP vers R2 (via R1 path)

  network 172.16.2.0 mask 255.255.255.0
  network 10.10.10.0 mask 255.255.255.0
```